

Wave 2 Temporal Model Entry Campaign Results

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report closes the approved `wave 2` temporal-model entry campaign. The campaign tested whether short temporal windows improve TE prediction beyond the closed static `wave 1` baselines.

The campaign completed all `9` planned runs with zero launcher failures. The campaign winner by the configured selection policy, minimum `test_mae` with `test_rmse` , `val_mae` , and parameter count as tie breakers, is `te_gru_sequence_remote_Fw` .

This result is a training-campaign winner. It is not yet an accepted `Track 2` model-verification winner, because the official `Track 2` matrix and visual reports still need to be refreshed with the new temporal candidates.

Campaign Artifacts

Artifact	Path
Campaign output	<code>output/training_campaigns/2026-05-24-11-20-37_wave2_temporal_model_entry_campaign_2026_05_24_11_01_15</code>
Campaign leaderboard	<code>output/training_campaigns/2026-05-24-11-20-37_wave2_temporal_model_entry_campaign_2026_05_24_11_01_15/campaign_leaderboard.yaml</code>
Best-run pointer	<code>output/training_campaigns/2026-05-24-11-20-37_wave2_temporal_model_entry_campaign_2026_05_24_11_01_15/campaign_best_run.yaml</code>
Execution report	<code>output/training_campaigns/2026-05-24-11-20-37_wave2_temporal_model_entry_campaign_2026_05_24_11_01_15/campaign_execution_report.md</code>
Planning report	<code>doc/reports/campaign_plans/wave2/2026-05-21-16-46-08_wave2_temporal_model_entry_campaign_plan_report.md</code>
Technical document	<code>doc/technical/2026-05/2026-05-21/2026-05-21-16-46-08_wave2_temporal_model_entry_plan.md</code>

Execution Summary

Field	Value
Campaign name	<code>wave2_temporal_model_entry_campaign_2026_05_24_11_01_15</code>
Started at	<code>2026-05-24T11:20:37</code>
Finished at	<code>2026-05-24T12:27:31</code>
Completed runs	<code>9</code>
Failed runs	<code>0</code>
Tested model types	<code>temporal_convolution</code> , <code>gru_sequence</code> , <code>lstm_sequence</code>
Tested direction scopes	<code>global</code> , <code>Fw</code> , <code>Bw</code>
Input contract	<code>centered sequence windows, length 33 , stride 4</code>

Campaign Winner

Field	Value
Run name	te_gru_sequence_remote_Fw
Run instance	2026-05-24-11-54-04__te_gru_sequence_remote_fw
Model family	gru_sequence_fw
Model type	gru_sequence
Direction scope	forward only
Trainable parameters	151,041
Validation MAE	0.003409
Validation RMSE	0.004010
Test MAE	0.003333
Test RMSE	0.003881

The winning checkpoint is stored at `output/training_runs/gru_sequence_fw/2026-05-24-11-54-04__te_gru_sequence_remote_fw/checkpoints/gru_sequence-epoch=045-val_mae=0.00340867.ckpt`.

Leaderboard

Rank	Run	Family	Scope	Test MAE	Test RMSE	Val MAE	Params
1	te_gru_sequence_remote_Fw	gru_sequence_fw	Fw	0.003333	0.003881	0.003409	151,041
2	te_lstm_sequence_remote_Fw	lstm_sequence_fw	Fw	0.003370	0.003921	0.003448	201,345
3	te_lstm_sequence_remote_global	lstm_sequence	global	0.003482	0.003948	0.003681	201,345
4	te_lstm_sequence_remote_Bw	lstm_sequence_bw	Bw	0.003557	0.004201	0.003815	201,345
5	te_gru_sequence_remote_global	gru_sequence	global	0.003591	0.004110	0.003707	151,041
6	te_temporal_convolution_sequence_remote_Fw	temporal_convolution_fw	Fw	0.003611	0.004183	0.003490	147,009
7	te_gru_sequence_remote_Bw	gru_sequence_bw	Bw	0.003631	0.004297	0.003867	151,041
8	te_temporal_convolution_sequence_remote_Bw	temporal_convolution_bw	Bw	0.003739	0.004369	0.003933	147,009
9	te_temporal_convolution_sequence_remote_global	temporal_convolution	global	0.003754	0.004266	0.003935	147,009

Technical Interpretation

The strongest temporal result is the forward-only GRU surface. It slightly outperforms the forward-only LSTM candidate in scalar test MAE while using about 25% fewer parameters. The GRU result is therefore the first temporal candidate to carry into the Track 2 visual refresh.

The LSTM family is still important: it is the best global temporal result and the best backward temporal result. The campaign does not show a universal dominance of one temporal family across all direction scopes.

The temporal convolution family is the smallest and simplest of the three families, but it ranks below both recurrent families in this first campaign. It remains useful as a lightweight local-context reference, not as the first promotion candidate.

The campaign does not displace the existing program-level tree_fw winner. The current registered program best remains te_hist_gbr_tabular_Fw_grid_depth6_lr008_leaf10 with test MAE 0.002743.

Registry Effects

Registry Scope	Current Family Best	Test MAE	Interpretation
temporal_convolution	te_temporal_convolution_sequence_remote_global	0.003754	New global temporal-convolution family best
temporal_convolution_fw	te_temporal_convolution_sequence_remote_Fw	0.003611	New forward temporal-convolution family best
temporal_convolution_bw	te_temporal_convolution_sequence_remote_Bw	0.003739	New backward temporal-convolution family best
gru_sequence	te_gru_sequence_remote_global	0.003591	New global GRU family best
gru_sequence_fw	te_gru_sequence_remote_Fw	0.003333	Campaign winner and strongest temporal scalar result
gru_sequence_bw	te_gru_sequence_remote_Bw	0.003631	New backward GRU family best
lstm_sequence	te_lstm_sequence_remote_global	0.003482	Best global temporal scalar result
lstm_sequence_fw	te_lstm_sequence_remote_Fw	0.003370	Second-best temporal scalar result
lstm_sequence_bw	te_lstm_sequence_remote_Bw	0.003557	Best backward temporal scalar result

Track 2 Boundary

Track 2 remains the official offline model-verification surface. The Wave 2 campaign provides trained candidates and scalar evidence, but it does not by itself establish official acceptance.

The following candidates should be prioritized in the Track 2 refresh:

- te_gru_sequence_remote_Fw as the campaign winner;
- te_lstm_sequence_remote_global as the best global temporal candidate;
- te_lstm_sequence_remote_Bw as the best backward temporal candidate;
- all remaining Wave 2 candidates as matrix rows if the refresh budget allows.

The official refresh must compare these candidates against the current tree, paper-derived, and Track 1 anchors on the same direction-aware curve reconstruction protocol.

Closeout Decision

The campaign is complete and successful from an execution standpoint: all 9 runs completed, the leaderboard and best-run artifacts exist, and the family registries were refreshed.

From a modeling standpoint, Wave 2 should remain open for official verification rather than immediate promotion. The GRU forward result is promising, while the LSTM global and backward results make the recurrent branch the useful temporal direction to inspect visually.

Recommended Follow-Up

1. Refresh the `Track 2` direction-aware matrix with the `9` Wave 2 temporal candidates.
2. Regenerate Track 2 best-model collages for the temporal winners.
3. Regenerate Track 2 multi-model overlays against `tree` , paper-derived, and Track 1 anchors.
4. Update the official Track 2 report ledger only after the refreshed matrix and visual PDFs have been validated.
5. Do not promote a temporal model to deployment planning until Track 2 visual review confirms curve-level benefit.